Introduction to Databases

CT042-3-1-IDB

Lecture 3: Data Model

Topic & Structure of The Lesson

- Data model
- Business rules
- Relational model
- Entity Relationship Model

Learning Outcomes

- At the end of this topic, You should be able to
  - Describe types of data model
  - Write business rules
  - Understand entity relationship model

Key Terms You Must Be Able To Use

- If you have mastered this topic, you should be able to use the following terms correctly in your assignments and exams:
  - Data model
  - Business rules
  - ERD

In this chapter, you will learn:

- Why data models are important
- About the basic data-modeling building blocks
- What business rules are and how they influence database design
- How the major data models evolved

The Importance of Data Models

- Data models
  - Relatively simple representations, usually graphical, of complex real-world data structures
  - Facilitate interaction among the designer, the applications programmer, and the end user

The Importance of Data Models (continued)

- End-users have different views and needs for data
- Data model organizes data for various users

Data Model Basic Building Blocks

- Entity - anything about which data are to be collected and stored
- Attribute - a characteristic of an entity
- Relationship - describes an association among entities
  - One-to-many (1:M) relationship
  - Many-to-many (M:N or M:M) relationship
  - One-to-one (1:1) relationship
- Constraint - a restriction placed on the data

Business Rules

- Brief, precise, and unambiguous descriptions of a policies, procedures, or principles within a specific organization
- Apply to any organization that stores and uses data to generate information
- Description of operations that help to create and enforce actions within that organization's environment

Business Rules (continued)

- Must be rendered in writing
- Must be kept up to date
- Sometimes are external to the organization
- Must be easy to understand and widely disseminated
- Describe characteristics of the data as viewed by the company

Discovering Business Rules

Sources of Business Rules:

- Company managers
- Policy makers
- Department managers
- Written documentation
  - Procedures
  - Standards
  - Operations manuals
- Direct interviews with end users

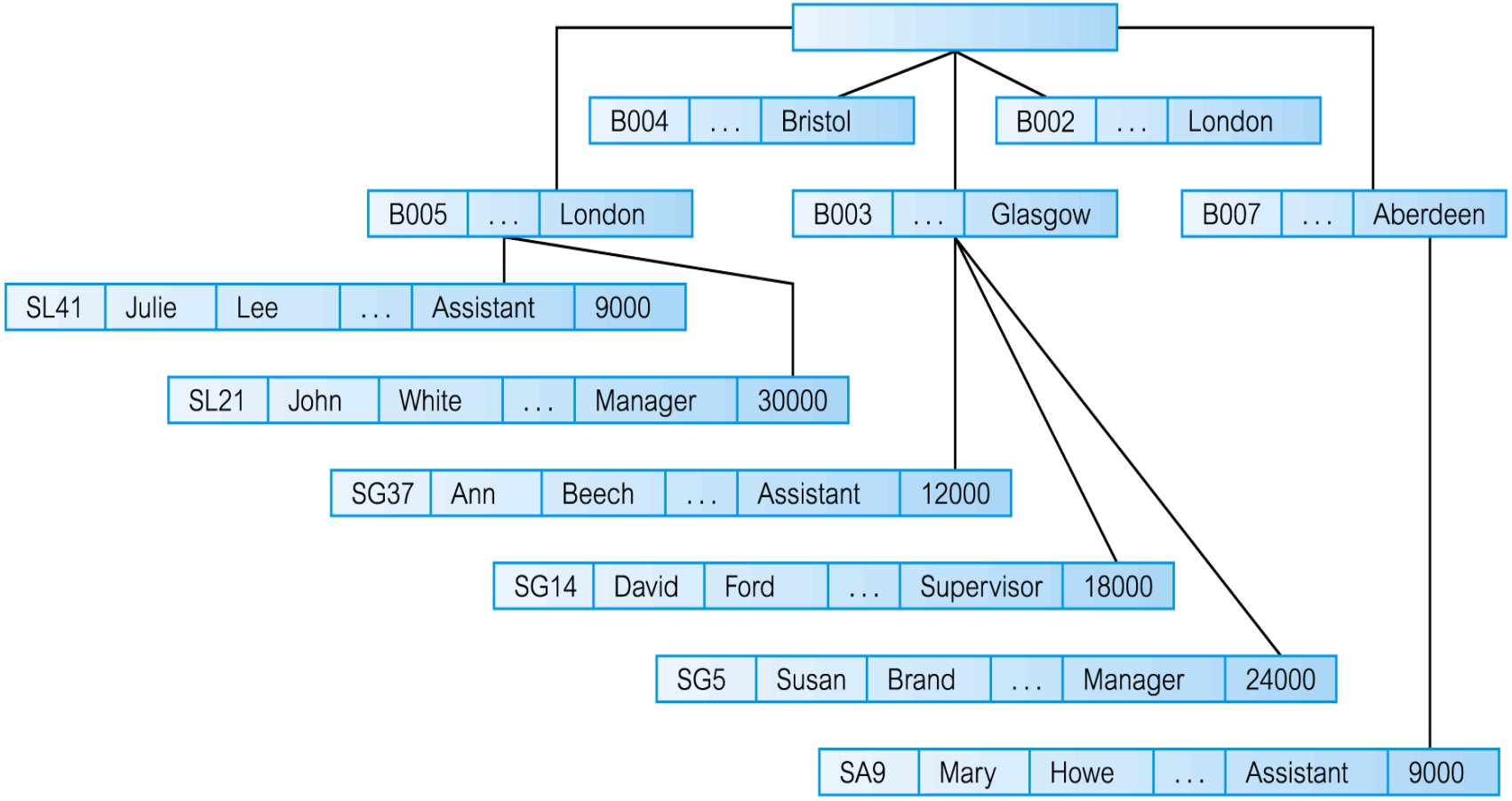
Discovering Business Rules (continued)

- Generally, nouns translate into entities
- Verbs translate into relationships among entities
- Relationships are bi-directional

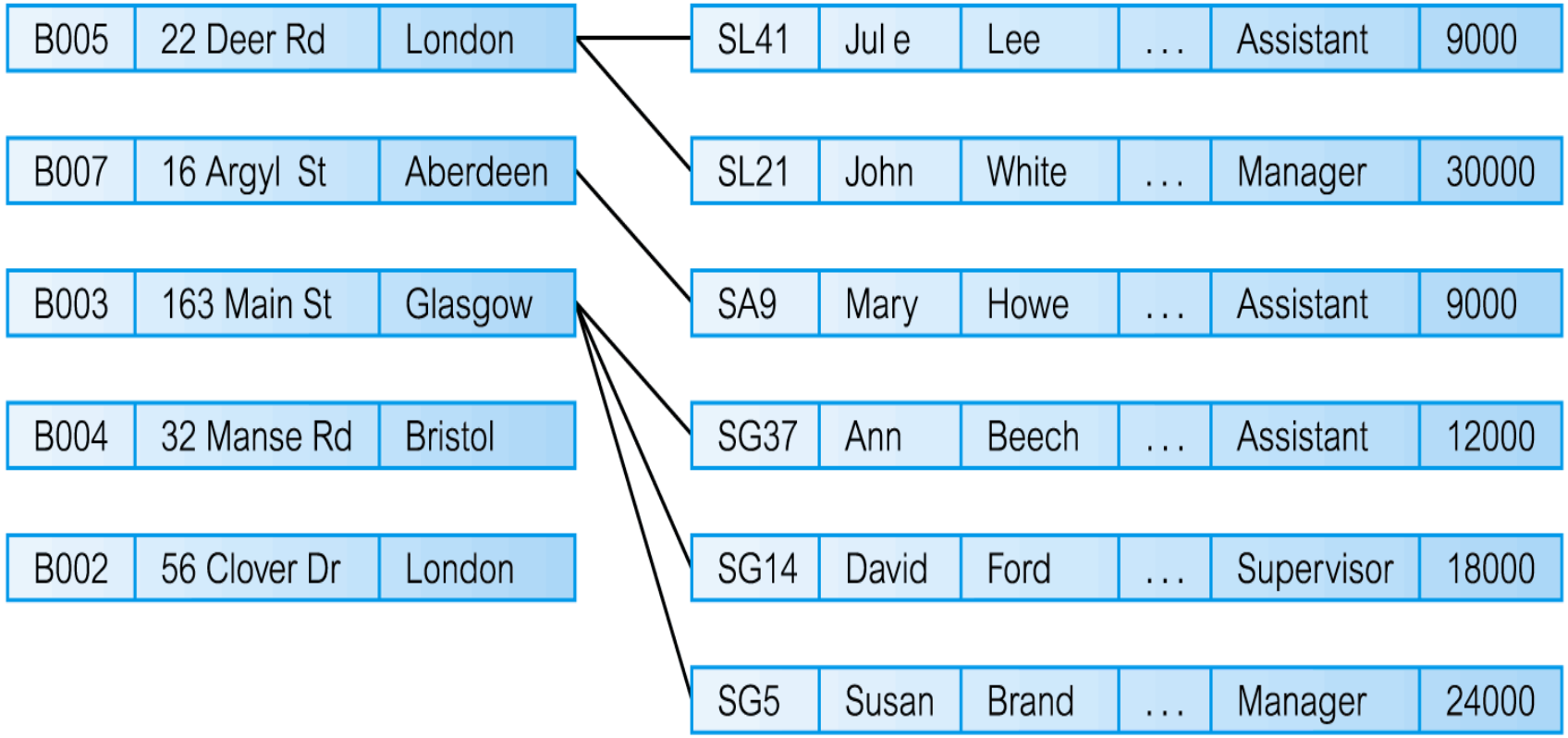
The Evolution of Data Models (continued)

- Hierarchical
- Network
- Relational
- Entity relationship
- Object Oriented

The Hierarchical Data Model



The Network Model



The Relational Model

- Developed by Codd (IBM) in 1970
- Considered ingenious but impractical in 1970
- Conceptually simple
- Computers lacked power to implement the relational model
- Today, microcomputers can run sophisticated relational database software

The Relational Model (continued)

- Table (relations)
  - Matrix consisting of a series of row/column intersections
  - Related to each other through sharing a common entity characteristic
- Relational diagram
  - Representation of relational database's entities, attributes within those entities, and relationships between those entities

The Relational Model (continued)

- Relational Table
  - Stores a collection of related entities
    - Resembles a file
- Relational table is purely logical structure
  - How data are physically stored in the database is of no concern to the user or the designer
  - This property became the source of a real database revolution

The Relational Model (continued)

FIGURE 2.3 Linking relational tables

Database name: Ch02_InsureCo Table name: AGENT (first six attributes)

AGENT_CODE	AGENT_LNAME	AGENT_FNAME	AGENT_INITIAL	AGENT_AREACODE	AGENT_PHONE
501	Alby	Alex	B	713	228-1249
502	Hahn	Leah	F	615	882-1244
503	Okon	John	T	615	123-5589

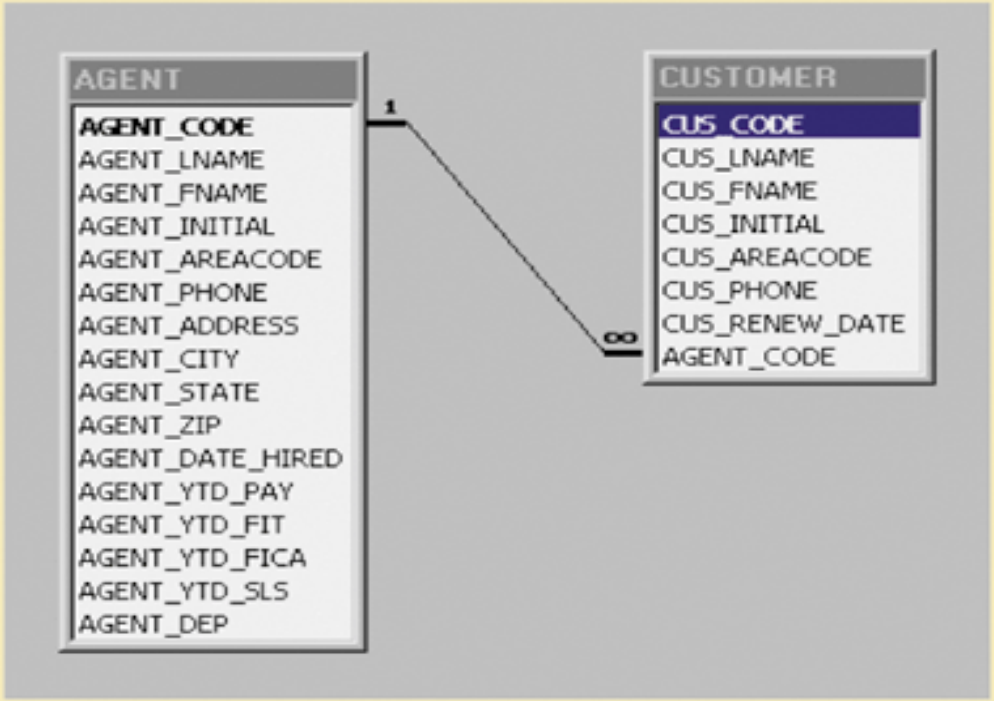
Link through AGENT_CODE

Table name: CUSTOMER

CUS_CODE	CUS_LNAME	CUS_FNAME	CUS_INITIAL	CUS_AREACODE	CUS_PHONE	CUS_RENEW_DATE	AGENT_CODE
10010	Ramas	Alfred	A	615	844-2573	05-Apr-2006	502
10011	Dunne	Leona	K	713	894-1238	16-Jun-2006	501
10012	Smith	Kathy	W	615	894-2285	29-Jan-2007	502
10013	Olowski	Paul	F	615	894-2180	14-Oct-2006	502
10014	Orlando	Myron		615	222-1672	28-Dec-2006	501
10015	O'Brian	Amy	B	713	442-3381	22-Sep-2006	503
10016	Brown	James	G	615	297-1228	25-Mar-2006	502
10017	Williams	George		615	290-2556	17-Jul-2006	503
10018	Farriss	Anne	G	713	382-7185	03-Dec-2006	501
10019	Smith	Olette	K	615	297-3809	14-Mar-2006	503

The Relational Model (continued)

FIGURE 2.4 A relational diagram



The Entity Relationship Model

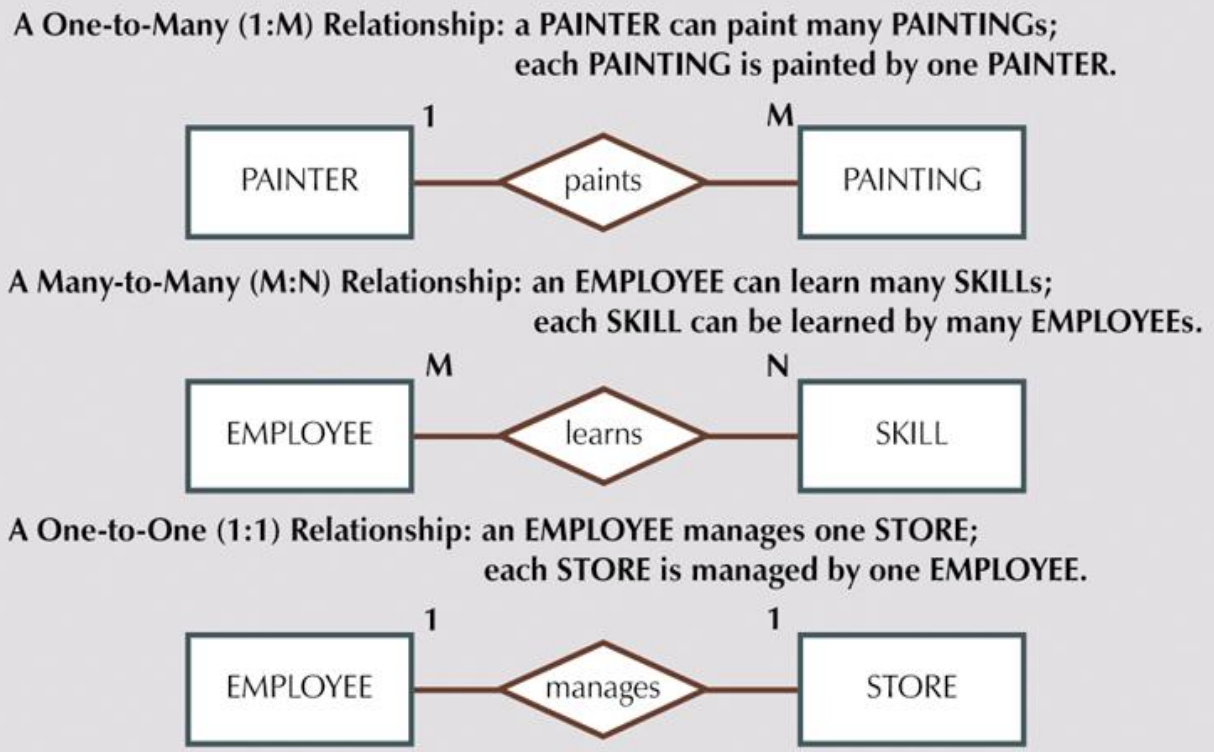
- Widely accepted and adapted graphical tool for data modeling
- Introduced by Chen in 1976
- Graphical representation of entities and their relationships in a database structure

The Entity Relationship Model (continued)

- Entity relationship diagram (ERD)
  - Uses graphic representations to model database components
  - Entity is mapped to a relational table
- Entity instance (or occurrence) is row in table
- Entity set is collection of like entities
- Connectivity labels types of relationships
  - Diamond connected to related entities through a relationship line

The Entity Relationship Model (continued)

FIGURE 2.5 The basic Chen ERD

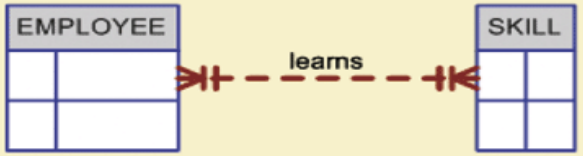


The Entity Relationship Model (continued)

FIGURE 2.6 The basic Crow's foot ERD

A One-to-Many (1:M) Relationship: a PAINTER can paint many PAINTINGs; each PAINTING is painted by one PAINTER.

A Many-to-Many (M:N) Relationship: an EMPLOYEE can learn many SKILLs; each SKILL can be learned by many EMPLOYEEs.



A One-to-One (1:1) Relationship: an EMPLOYEE manages one STORE; each STORE is managed by one EMPLOYEE.

Quick Review Question

- What is data model?
- What is the source of business rules?
- What is entity relationship model?

Summary of Main Teaching Points

- Data models are simple representations, usually graphical, of complex real-world data structures
- Business rules are brief, precise, and unambiguous descriptions of a policies, procedures, or principles within a specific organization.
- Entity relationship models are graphical representation of entities and their relationships in a database structure

Question and Answer Session

Q & A

What we will cover next

- Characteristics of relational table
- Different types of keys
- Integrity rules
- Data dictionary